FNCE 30007 DERIVATIVE SECURITIES

Tutorial 4 Questions with solutions: Introduction to options

Question 1

What is a lower bound for the price of a six-month call option on a non-dividend-paying stock when the stock price is $80, the strike price is $75, and the continuously compounded risk-free interest rate is 10% per annum?

Answer

We know that a portfolio of one call and the PV of the strike is at least as good as the stock. It follows that the value of a call plus the PV of the strike is at least as valuable as the stock and therefore:

$$C \geq S_0 - Ke^{-rT}$$
$$C \geq 80 - 75 \times e^{-0.1\times 0.5}$$
$$C \geq 8.66.$$

A lower bound for the call is $8.66

Question 2

What is a lower bound for the price of a two-month European put option on a non-dividend-paying stock when the stock price is $58, the strike price is $65, and the risk-free interest rate is 5% per annum?

Answer

Analogous arguments to Question 1, a portfolio consisting of one put plus one share has a payoff at least as good as the depositing the PV of the strike today. Therefore:

$$p \geq Ke^{-rT} - S_0$$
$$p \geq 65e^{-0.05/6} - 58$$
$$p \geq 6.46$$

A lower bound for the put is $6.46.

Question 3

Calls were traded on exchanges before puts. During this period of time when calls were traded but puts were not, how would you create a European put option on a non-dividend-paying stock synthetically? Assume no dividends are paid during the life of the contracts.

Answer

Put-call parity can be used to create a put from a portfolio call and a position in stock and cash markets. The payoff of a portfolio of a put plus the stock equals the payoff of a portfolio of a call plus the present value of the strike price invested in cash, assuming the call and the put have the same strike price and maturity date.

We can create a put by buying the call, shorting the stock, and keeping an amount of cash that when invested at the risk-free rate will grow to be sufficient to exercise the call.
If the stock price is above the strike price, the call is exercised using cash from the investment, and the stock is delivered to close out the short position for no net payoff.
If the stock price is below the strike price, the call is not exercised and the short position is closed out for a gain equal to the payoff $K - S_T$*, which is that of a put with a strike price of* K.

Question 4

The price of a European call that expires in six months and has a strike price of $30 is $2. The underlying stock price is $29, and a dividend of $0.50 is expected in two months and again in five months. Risk-free interest rates for all maturities are 10%. What is the price of a European put option that expires in six months and has a strike price of $30?

Answer

We know that the following two portfolios have the same payoffs at expiry:

1. *A European call plus cash equal to* $D + Ke^{-rT}$ *that earns the risk-free rate*
2. *A European put plus one share*

where D *is the present value of the dividend.*

Therefore the options must satisfy the put-call parity condition:

$$C + D + Ke^{-rT} = p + S_0.$$

Solving for p*:*

$$p = C + D + Ke^{-rT} - S_0$$
$$p = 2 + 0.5 \times (e^{-0.1/6} + e^{-0.1\times 5/12}) + 30e^{-0.1/2} - 29$$
$$p = 2.51.$$

Question 5

Explain carefully the arbitrage opportunities in Question 4 if the European put price is $3.

Answer

In this case, the put is too valuable. An arbitrageur can exploit this by creating a portfolio consisting of:

1. *Long one call*
2. *Short one put*
3. *Short one stock*

This generates cash of $-2 + 3 + 29 = 30$ *today which is invested at the risk-free rate. The short stock position generates a negative cash flow of $0.50 for each dividend which must be funded from our cash investment.*

To show this is an arbitrage, consider the total cash flows when $S_T > K$, $S_T < K$ *and* $S_T = K$ *at expiry:*

Regardless of the spot price at expiry, the terminal value of the cash investment is:

$$(30 - 0.5e^{-0.1/6} - 0.5e^{-0.1\times 5/12})e^{0.1/2} = 30.52$$

The arbitrageur must return the stock to the lender at expiry. This costs the arbitrageur $30 *in all three scenarios:*

- *If* $S_T > 30$*, the put expires worthless and the arbitrageur exercises the call to buy and deliver the stock at a cost of $30, leaving PnL of $0.52.*

- *If* $S_T < K$*, the put is exercised, which gives the arbitrageur one unit of the stock at a cost of $30, while the call expires worthless, leaving the same PnL for the arbitrageur.*

- *If* $S_T = K$*, assume the call and put are not exercised (it makes no difference if either are) and the stock is bought on the cash market for $30, leaving the same PnL for the arbitrageur.*

Note that the present value of the PnL is $0.52 \times e^{-0.05} = 0.49$ *which corresponds to the PnL calculation in the textbook.*

Question 6

Give an intuitive explanation of why the early exercise of an American put often becomes more attractive as the risk-free rate increases and volatility decreases.

Answer

Early exercise allows the option holder to begin earning interest on the strike price before option maturity. The value of early exercises increases when the interest earned on the strike price is higher.

For early exercise to be optimal, the interest benefit must exceed the time-value embedded in the option if it were not exercised. This time-value is a function of volatility – the larger is the volatility, the larger is the potential upside of the option relative to the downside (in most cases).

Note that the textbook answer discusses this in relation to the concept of "insurance value" instead of time-value. This term makes sense for option owners who are hedging an underlying exposure but not to other traders who may take a position in the put to speculate.

Question 7

After reading the GameStop case, think about the following questions before the tutorial.

7.1 How Would You Trade GameStop?
POV: It is late 2020. You believe that GameStop's turnaround strategy could gain momentum and that its share price may rise substantially. However, given the uncertainty surrounding the company, you do not want to risk losing the full amount required to purchase GME shares.

Which of the following option positions (Long Call, Long Put, Short Put, Short Call) would best reflect your market view and risk preference? Explain your choice.

Answer: *Long Call*

A long call is appropriate because the investor is bullish on GME and expects the share price to rise substantially. It provides upside exposure if the share price increases, while the maximum possible loss is limited to the option premium paid. This makes a long call particularly suitable given the high uncertainty surrounding GME, as the investor can benefit from a large upside move without committing the full amount required to purchase the shares.

7.2 Put–Call Parity in Current Market Conditions

For this question, use the current (August 2026) market data provided in the second worksheet (Tutorial - GME 2026) of GameStop_Options_Case_FNCE30007.xlsx.

(a) Using the data provided, calculate the lower and upper bounds for the American put option implied by the American put-call parity relationship.

(b) Compare the observed market put price with your calculated range. Does the observed put price fall within the no-arbitrage range?

(c) Compare your result with the January 2021 GameStop short-squeeze example discussed in the lecture. What do you notice, and what might explain the difference?

Answer:
For a non-dividend-paying stock, the American put price should satisfy:
$C+Ke^{-rT}-S0 < P < C+K-S0$
Using the market data provided in Tutorial - GME 2026:

(a) Lower Bound：$C+Ke^{-rT}-S0=\$1.94$

(b) Upper Bound：$C+K-S0=\$2.07$
Therefore, the American put no-arbitrage range is: $\$1.94<P<\$2.07$

***(c)** The observed market put price is $P=\$2.00$*
Since: $\$1.94<\$2.00<\$2.07$
The observed put price falls within the American put no-arbitrage range for 2026 data.
A potential explanation is the difference in market conditions. During the 2021 short squeeze, GME experienced extreme volatility, severe short-selling constraints and substantial trading frictions. These conditions made arbitrage more difficult and costly, allowing larger pricing inconsistencies to persist.
By contrast, the 2026 market conditions are relatively normal. Arbitrage and short-selling mechanisms can operate more effectively, helping keep the prices of the stock, calls and puts more closely aligned with no-arbitrage relationships.